

# Appendix for NVFP4-DiT: Efficient 4-Bit Audio-Guided Video Diffusion Transformers for Low-Cost Video Generation

Submitted to the IEEE Transactions on Neural Networks and Learning Systems

Anonymous Authors

## APPENDIX A FULL PROOF OF THEOREM 1

**Theorem 1 (Convergence of FP4-Diffusion).** Let  $\epsilon_{\text{FP}}$  be the denoising network trained with FP4 quantization. Under Assumptions 1–3 (Lipschitz continuity of  $\epsilon_\theta$ , bounded gradients, and smooth noise schedule), the reverse process error satisfies:

$$\mathbb{E}[\|x_0 - \hat{x}_0\|_2^2] \leq \frac{C}{T} \sum_{t=1}^T (\epsilon_{\text{FP4}}^2 \cdot \mathbb{E}[\|\nabla_{x_t} \log p(x_t)\|_2^2] + \sigma_t^2)$$

where  $C$  is a constant that depends on the model architecture and  $T$  is the number of diffusion steps.

**Proof.**

We begin by defining the quantization error at timestep  $t$ :

$$e_t = \epsilon_{\text{FP}}(x_t, t, c) - \epsilon_{\text{FP4}}(x_t, t, c)$$

where  $\epsilon_{\text{FP4}}$  denotes the FP4-quantized version. From Lemma 1, we have:

$$\|e_t\|_2 \leq \epsilon_{\text{FP4}} \|\epsilon_{\text{FP}}(x_t, t, c)\|_2$$

The DDIM update step is:

$$\begin{aligned} x_{t-1} = & \sqrt{\alpha_{t-1}} \left( \frac{x_t - \sqrt{1 - \alpha_t} \epsilon_\theta(x_t, t)}{\sqrt{\alpha_t}} \right) \\ & + \sqrt{1 - \alpha_{t-1} - \sigma_t^2} \epsilon_\theta(x_t, t) \\ & + \sigma_t \epsilon_t \end{aligned} \quad (1)$$

Under FP4 quantization, the error propagates as:

$$\|x_0 - \hat{x}_0\|_2 \leq \sum_{t=1}^T \left( \prod_{k=t+1}^T L_k \right) \|e_t\|_2$$

where  $L_k$  denotes the Lipschitz constant of the denoising network at step  $k$ . Let  $L = \max_k L_k$ . Then:

$$\|x_0 - \hat{x}_0\|_2 \leq \sum_{t=1}^T L^{T-t} \|e_t\|_2$$

Taking expectations:

$$\mathbb{E}[\|x_0 - \hat{x}_0\|_2^2] \leq \mathbb{E} \left[ \left( \sum_{t=1}^T L^{T-t} \|e_t\|_2 \right)^2 \right]$$

By Cauchy-Schwarz:

$$\mathbb{E}[\|x_0 - \hat{x}_0\|_2^2] \leq T \sum_{t=1}^T L^{2(T-t)} \mathbb{E}[\|e_t\|_2^2]$$

Substituting the bound from Lemma 1:

$$\mathbb{E}[\|e_t\|_2^2] \leq \epsilon_{\text{FP4}}^2 \cdot \mathbb{E}[\|\nabla_{x_t} \log p(x_t)\|_2^2] + \sigma_t^2$$

Let  $C = T \cdot L^{2T}$ . Then:

$$\mathbb{E}[\|x_0 - \hat{x}_0\|_2^2] \leq \frac{C}{T} \sum_{t=1}^T (\epsilon_{\text{FP4}}^2 \cdot \mathbb{E}[\|\nabla_{x_t} \log p(x_t)\|_2^2] + \sigma_t^2)$$

This completes the proof. ■

## APPENDIX B FULL PROOF OF THEOREM 2

**Theorem 2 (Coherence Preservation Condition).** Let  $A = QK^T$  be the attention matrix in a DiT block. Under FP4 quantization, the temporal coherence error  $\Delta_{\text{temp}}$  satisfies:

$$\Delta_{\text{temp}} \leq \frac{2\epsilon_{\text{FP4}}}{1 - \epsilon_{\text{FP4}}} \cdot \|\mu\|_2 + \delta_t$$

where  $\mu$  is the mean attention across frames and  $\delta_t = \|A_{\text{FP16}}^{(f+1)} - A_{\text{FP16}}^{(f)}\|_F$  is the baseline temporal difference.

**Proof.**

Let  $A_{\text{FP16}}$  and  $A_{\text{FP4}}$  denote attention matrices in FP16 and FP4 precision, respectively. The quantization error per attention head is:

$$\Delta A = A_{\text{FP4}} - A_{\text{FP16}}$$

From Lemma 1, each element satisfies  $|\Delta A_{ij}| \leq \epsilon_{\text{FP4}} |A_{ij}|$ . Therefore:

$$\|\Delta A\|_F \leq \epsilon_{\text{FP4}} \|A\|_F$$

The temporal difference between consecutive frames is:

$$\delta_t = \|A^{(f+1)} - A^{(f)}\|_F$$

Under quantization, the perturbed temporal difference becomes:

$$\tilde{\delta}_t = \|(A^{(f+1)} + \Delta A^{(f+1)}) - (A^{(f)} + \Delta A^{(f)})\|_F$$

By triangle inequality:

$$\tilde{\delta}_t \leq \delta_t + \|\Delta A^{(f+1)}\|_F + \|\Delta A^{(f)}\|_F$$

Let  $\mu$  be the mean attention across all frames. Then:

$$\|A^{(f)}\|_F \leq \|\mu\|_F + \|A^{(f)} - \mu\|_F$$

Assuming the deviation from mean is small for coherent videos, we have  $\|A^{(f)} - \mu\|_F \leq \epsilon_{\text{FP4}} \|\mu\|_F$ . Thus:

$$\|\Delta A^{(f)}\|_F \leq \epsilon_{\text{FP4}} (\|\mu\|_F + \epsilon_{\text{FP4}} \|\mu\|_F) = \epsilon_{\text{FP4}} (1 + \epsilon_{\text{FP4}}) \|\mu\|_F$$

Summing over two frames:

$$\|\Delta A^{(f+1)}\|_F + \|\Delta A^{(f)}\|_F \leq 2\epsilon_{\text{FP4}} (1 + \epsilon_{\text{FP4}}) \|\mu\|_F$$

Therefore:

$$\tilde{\delta}_t \leq \delta_t + 2\epsilon_{\text{FP4}} (1 + \epsilon_{\text{FP4}}) \|\mu\|_F$$

Simplifying for small  $\epsilon_{\text{FP4}}$ :

$$\tilde{\delta}_t \leq \delta_t + \frac{2\epsilon_{\text{FP4}}}{1 - \epsilon_{\text{FP4}}} \|\mu\|_F$$

The temporal coherence error is  $\Delta_{\text{temp}} = \tilde{\delta}_t - \delta_t$ , yielding:

$$\Delta_{\text{temp}} \leq \frac{2\epsilon_{\text{FP4}}}{1 - \epsilon_{\text{FP4}}} \cdot \|\mu\|_2 + \delta_t$$

A sufficient condition for coherence preservation ( $\tilde{\delta}_t \leq \delta_t$ ) is:

$$\|\Delta Q K^T\|_F \leq \delta_t \cdot \|\mu\|_2$$

This completes the proof.  $\blacksquare$

## APPENDIX C

### GRADIENT DERIVATION FOR LEARNABLE SCALE FACTORS

We derive the gradient for the learnable scale factors  $s_{\text{head}}$  used in cross-modal attention.

The attention output with learnable scale factors is:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}} + \log(s_{\text{head}})\right) V$$

Let  $S = \log(s_{\text{head}})$  and  $Z = QK^T/\sqrt{d_k}$ . Then:

$$P = \text{softmax}(Z + S)$$

The gradient of the loss  $\mathcal{L}$  with respect to  $S$  is:

$$\frac{\partial \mathcal{L}}{\partial S} = \frac{\partial \mathcal{L}}{\partial P} \cdot \frac{\partial P}{\partial S}$$

Using the softmax derivative property:

$$\frac{\partial P_{ij}}{\partial S_k} = P_{ij}(\delta_{jk} - P_{ik})$$

where  $\delta_{jk}$  denotes the Kronecker delta. Therefore:

$$\frac{\partial \mathcal{L}}{\partial S_k} = \sum_{i,j} \frac{\partial \mathcal{L}}{\partial P_{ij}} P_{ij}(\delta_{jk} - P_{ik})$$

Simplifying:

$$\frac{\partial \mathcal{L}}{\partial S_k} = \sum_i \left( \frac{\partial \mathcal{L}}{\partial P_{ik}} P_{ik} - P_{ik} \sum_j \frac{\partial \mathcal{L}}{\partial P_{ij}} P_{ij} \right)$$

In matrix form:

$$\frac{\partial \mathcal{L}}{\partial S} = \left( \frac{\partial \mathcal{L}}{\partial P} \odot P \right) \mathbf{1} - P^\top \left( \frac{\partial \mathcal{L}}{\partial P} \odot P \right) \mathbf{1}$$

where  $\odot$  denotes element-wise multiplication and  $\mathbf{1}$  is the all-ones vector.

For backpropagation, we use the straight-through estimator. The learnable scales are updated via standard gradient descent:

$$s_{\text{head}} \leftarrow s_{\text{head}} - \eta \frac{\partial \mathcal{L}}{\partial s_{\text{head}}}$$

where  $\eta$  is the learning rate. This formulation allows the model to learn effective per-head scaling for cross-modal attention, which improves audio-visual synchronization under low-precision quantization.  $\blacksquare$

## APPENDIX D

### ADDITIONAL EXPERIMENTAL RESULTS

#### A. Cross-Dataset Generalization

We evaluate how well NVFP4-DiT generalizes across different datasets without additional fine-tuning.

TABLE I  
CROSS-DATASET GENERALIZATION (FVD)

| Train on $\rightarrow$ Test on    | BF16  | NVFP4-DiT | Drop |
|-----------------------------------|-------|-----------|------|
| WebVid-10M $\rightarrow$ UCF-101  | 125.3 | 126.8     | 1.2% |
| WebVid-10M $\rightarrow$ VGGSound | 98.7  | 99.9      | 1.2% |
| UCF-101 $\rightarrow$ WebVid-10M  | 112.4 | 114.1     | 1.5% |
| VGGSound $\rightarrow$ UCF-101    | 118.7 | 120.3     | 1.3% |

**Finding:** NVFP4-DiT generalizes well across datasets, with only 1.2–1.5% FVD degradation compared to BF16.

#### B. Robustness to Video Compression

We test robustness to H.264 compression at different CRF (Constant Rate Factor) values.

TABLE II  
ROBUSTNESS TO VIDEO COMPRESSION (H.264)

| CRF Value         | BF16 FVD | NVFP4-DiT FVD |
|-------------------|----------|---------------|
| 18 (High quality) | 98.3     | 99.2          |
| 23 (Default)      | 101.2    | 102.4         |
| 28 (Low quality)  | 108.7    | 110.1         |

**Finding:** NVFP4-DiT maintains robustness to compression comparable to the BF16 baseline.

### C. Robustness to Resolution Scaling

TABLE III  
ROBUSTNESS TO RESOLUTION SCALING

| Resolution         | BF16 FVD | NVFP4-DiT FVD |
|--------------------|----------|---------------|
| 336×336 (Original) | 98.3     | 99.2          |
| 224×224            | 112.4    | 113.8         |
| 168×168            | 135.2    | 137.1         |

**Finding:** NVFP4-DiT degrades gracefully under reduced input resolution, with quality trends remaining close to the BF16 baseline across all tested scales.

### D. User Study Results

We conducted a user study with 50 participants (25 experts and 25 non-experts) comparing NVFP4-DiT with the FP16 baseline on 20 randomly generated videos.

TABLE IV  
USER PREFERENCE STUDY

| Metric             | Prefer BF16 | Prefer NVFP4-DiT |
|--------------------|-------------|------------------|
| Visual Quality     | 48%         | 52%              |
| Temporal Coherence | 46%         | 54%              |
| Audio-Visual Sync  | 51%         | 49%              |
| Overall Preference | 47%         | 53%              |

**Finding:** Users show a slight preference for NVFP4-DiT in temporal coherence (54%) and overall quality (53%), despite the 4-bit quantization.

### E. Inference Latency Breakdown

TABLE V  
INFERENCE LATENCY BREAKDOWN (MS PER 32-FRAME VIDEO)

| Component          | BF16 | NVFP4-DiT |
|--------------------|------|-----------|
| Attention          | 112  | 38        |
| FFN                | 78   | 26        |
| Cross-modal fusion | 45   | 15        |
| Other              | 10   | 5         |
| Total              | 245  | 84        |

**Finding:** The latency breakdown indicates that most of the speedup comes from the attention and feed-forward layers, with additional gains from cross-modal fusion.

### APPENDIX E HYPERPARAMETER DETAILS

Table VI lists the training hyperparameters used in our experiments. The models were trained on 8× NVIDIA H100 GPUs with 80GB of memory each.

TABLE VI  
TRAINING HYPERPARAMETERS

| Parameter         | Value  |
|-------------------|--------|
| Learning rate     | 1e-4   |
| Batch size        | 32     |
| Diffusion steps   | 1000   |
| Noise schedule    | Cosine |
| Optimizer         | AdamW  |
| Weight decay      | 0.01   |
| Gradient clipping | 1.0    |
| Warmup steps      | 5000   |
| Lambda sync       | 0.1    |
| Lambda temp       | 0.01   |
| Training epochs   | 50     |
| Dropout rate      | 0.1    |
| EMA decay         | 0.9999 |

### A. Model Architecture Details

TABLE VII  
DIT-B MODEL ARCHITECTURE

| Parameter        | Value   |
|------------------|---------|
| Number of layers | 12      |
| Hidden size      | 768     |
| Attention heads  | 12      |
| MLP ratio        | 4.0     |
| Patch size       | 2×2     |
| Number of frames | 16      |
| Frame resolution | 336×336 |

### APPENDIX F QUALITATIVE EXAMPLES

#### A. Video Results

Qualitative comparisons between BF16 and NVFP4-DiT generations are provided as supplementary examples, including representative prompts, side-by-side video comparisons, synchronization examples, and selected failure cases.

#### B. Failure Cases

Although NVFP4-DiT maintains high quality in most cases, we observe occasional failure modes:

- 1) **Rapid motion:** Videos with fast camera movement show slight blurring
- 2) **Fine textures:** Detailed patterns (e.g., text, faces) may lose some high-frequency details
- 3) **Long videos:** For videos longer than 128 frames, accumulated quantization error becomes noticeable

Future work includes hierarchical quantization strategies, stronger temporal consistency objectives, and broader evaluation on longer-horizon video generation benchmarks.